

The Derivative of the Sigmoid Function

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Leads to:** Logistic regression ([Note 75](#)); Gradient boosting ([Note 120](#)); Multi-layer perceptron (MLP) ([Note 1003](#)); Backpropagation ([Note 1015](#)); Vanishing gradient ([Note 1018](#)).
- **Compare with:** Tanh ([Note 1027](#)).

1. Overview

Key point: The derivative of the sigmoid can be written using the sigmoid itself: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.

Gradient descent on the log loss (the next Note) needs the derivative of the sigmoid function. The same derivative appears again in neural networks, where a neuron can use the sigmoid to produce its output. This short Note derives it once.

As a reminder, the sigmoid is

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid squeezes any number into the range 0 to 1 ([sigmoid Note](#)).

2. Two rules we need

Key point: The chain rule, and the derivative of e^{-z} , which is $-e^{-z}$.

Chain rule: to differentiate a function of a function, differentiate the outer one, keep the inside, and multiply by the derivative of the inside. For a reciprocal:

$$\frac{d}{dz} \left(\frac{1}{u} \right) = -\frac{1}{u^2} \cdot \frac{du}{dz}$$

Exponential: the derivative of e^z is e^z itself. By the chain rule, the derivative of e^{-z} is e^{-z} times the derivative of $-z$, which is -1 :

$$\frac{d}{dz} e^{-z} = -e^{-z}$$

3. The derivation

Key point: Differentiate, then split the fraction into $\sigma(z)$ times $(1 - \sigma(z))$.

3.1 Differentiate

Key point: Treat $\sigma(z)$ as $1/u$ with $u = 1 + e^{-z}$.

Write $u = 1 + e^{-z}$, so $\sigma(z) = 1/u$. The derivative of u is $0 + (-e^{-z}) = -e^{-z}$. Using the reciprocal rule:

$$\sigma'(z) = -\frac{1}{(1 + e^{-z})^2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

The two minus signs cancel.

3.2 Split into two fractions

Key point: One fraction is $\sigma(z)$ itself; the other turns out to be $1 - \sigma(z)$.

Split the result into a product of two fractions:

$$\sigma'(z) = \frac{1}{1 + e^{-z}} \cdot \frac{e^{-z}}{1 + e^{-z}}$$

The first fraction is $\sigma(z)$. For the second, add and subtract 1 in the numerator:

$$\frac{e^{-z}}{1 + e^{-z}} = \frac{(1 + e^{-z}) - 1}{1 + e^{-z}} = 1 - \frac{1}{1 + e^{-z}} = 1 - \sigma(z)$$

3.3 The result

Key point: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.

Putting the two pieces together:

$$\sigma'(z) = \sigma(z) (1 - \sigma(z))$$

With numbers: at $z = 2$, $\sigma(2) = 0.88$, so $\sigma'(2) = 0.88 \times 0.12 = 0.105$. Figure 1 (Section 4) draws the two factors as the blue and orange parts of a bar of height 1.

The form $\sigma(1 - \sigma)$ is very convenient. When a model has already computed $\sigma(z)$ for its prediction, the derivative costs just one subtraction and one multiplication.

4. The shape of the derivative

Key point: The derivative is largest, 0.25, at $z = 0$ and falls towards 0 on both sides.

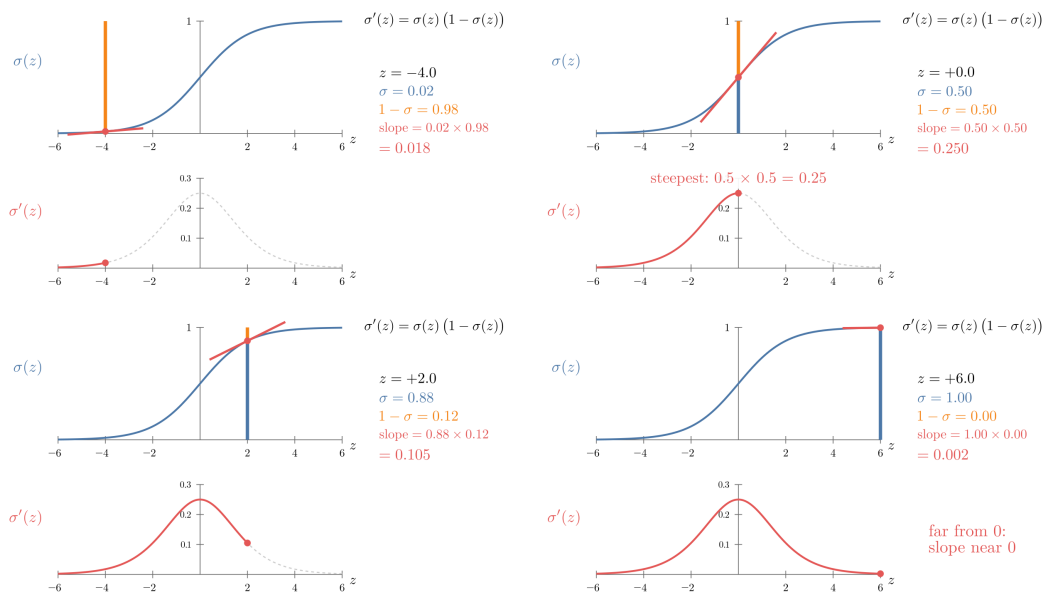


Figure 1: A tangent line (red) rides the sigmoid; its slope is traced below. The bar at z splits into $\sigma(z)$ (blue) and $1 - \sigma(z)$ (orange), and the slope is their product. Key frames: $z = -4, 0, 2, 6$

In Figure 1, watch the two coloured parts of the bar: the slope is large only when both parts are large, which happens at $z = 0$, where each is 0.5. Reading the derivative as the slope of a tangent line that moves along the curve follows Sanderson's *Essence of Calculus* (3Blue1Brown, "The paradox of the derivative").

z	-4	-2	0	2	4
$\sigma(z)$	0.018	0.119	0.5	0.881	0.982
$\sigma'(z)$	0.018	0.105	0.25	0.105	0.018

- At $z = 0$ the sigmoid is steepest: $0.5 \times 0.5 = 0.25$. The value 0.25 is the largest value the derivative can take.
- Far from 0, the sigmoid is almost flat, near 0 or 1, so its slope is almost 0.
- The curve is symmetric: $\sigma'(-z) = \sigma'(z)$.

Python: The derivative, checked numerically.

```
def sigmoid(z):
    return 1 / (1 + np.exp(-z))
```

```
def sigmoid_derivative(z):
```

```
    return sigmoid(z) * (1 - sigmoid(z))

h = 1e-5
(sigmoid(2 + h) - sigmoid(2 - h)) / (2 * h)    # 0.104994
sigmoid_derivative(2)                            # 0.104994
The numerical slope (rise over a tiny run) matches the formula to six decimals.
```

Extra: Because the derivative is at most 0.25, and almost 0 for large or small z , stacking many sigmoid layers in a deep neural network multiplies many small numbers together. The gradients become tiny and learning stalls: the **vanishing gradient problem**. Saturating sigmoids make gradient-based learning very difficult, so the sigmoid is now discouraged inside the hidden layers of a network, and ReLU is the default choice (Goodfellow et al. §6.1, §6.3.2).

5. Summary

- $\sigma(z) = 1/(1 + e^{-z})$.
- $\sigma'(z) = \sigma(z)(1 - \sigma(z))$, found with the chain rule and one algebra step.
- The derivative peaks at 0.25 when $z = 0$ and approaches 0 far from it.
- The next Note uses it to derive the gradient of the log loss.

6. Sources

Built from

- CampusX, “Derivative of Sigmoid Function”, YouTube, <https://www.youtube.com/watch?v=awjXaFR>
- Sanderson, G. (3Blue1Brown), “The paradox of the derivative”, *Essence of Calculus*, chapter 2. 3blue1brown.com/lessons/derivatives

Other references

- **Goodfellow et al.:** Goodfellow, I., Bengio, Y. and Courville, A. *Deep Learning*. MIT Press, 2016. Sections 6.1 and 6.3.2.

7. Key terms

Term	Meaning
Derivative	The rate of change, or slope, of a function at a point
Chain rule	To differentiate a function of a function, multiply the outer derivative by the inner derivative
Vanishing gradient	Gradients shrinking towards 0 as they pass through many layers, which slows learning